

Persistent environment

Event-triggered state snapshot
arrival / completion at t_k

10-robot persistent fleet
positions, loads, ownership

Queue and shared resources
four stairs, handover regions

Event-driven masked SMDP-PPO

284-D state
encoder

Up to 1536
joint candidates

mask

Context-candidate
actor-critic

Candidate construction and feasibility

Legal modes

SINGLE-CAR · SINGLE-DOG
CAR-DOG-CAR

Joint candidate

task × robots × stair
mode-feasible enumeration

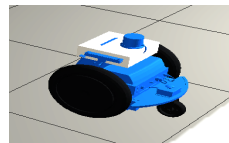
Decision context

ETA · geometry · queue
resources · handover risk

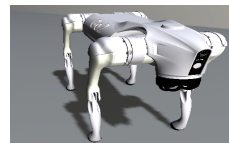
Execution interface

Reserve assignment and execute
SCAN-Planner / robot skills

Mode-conditioned robot skills



Carter
wheeled; level floor



Go2
legged; stair-capable

Next-event update: positions, ownership, queue, and released reservations